

SRP

Python <
 compiler
 interpreter : (python.exe)

command prompt - python

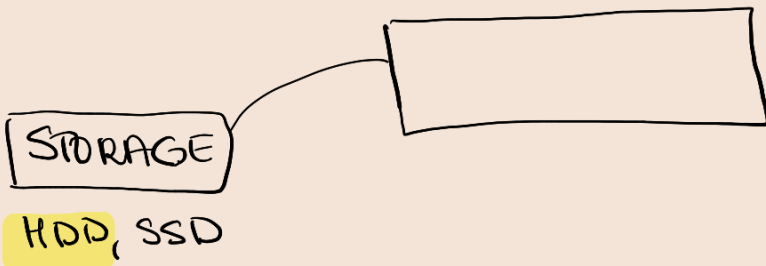
Write Python directly in terminal : PATH $\rightarrow$ environment variable

OS $\rightarrow$ everything is a file. $\rightarrow$ File = a finite succession of bytes

Measure for memory : BYTE

$\rightarrow$ FILE SYSTEM

Windows File Format



FAT

NTFS

trees $\rightarrow$ always use the absolute path in a project, exams!
relative path

Every file can be identified by its PATH.

variable - tuple $\leftarrow$ label (unique) $\rightarrow$ can't start with a digit

In Python, the input is always a string!

mantissa

type(sth) $\rightarrow$ prints the datatype of "sth".

Can $\sqrt{2}$, pi be stored in the memory? No $\rightarrow \infty$ decimals.

1)

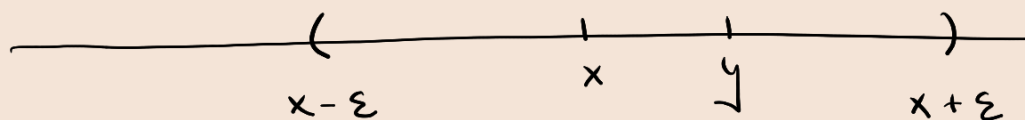
```
sth = input("Give m: ")  
print(sth)
```

 $\rightarrow$ string (by default)

type(sth)

2) `sth = int(input("Give m:"))`
`print(sth)`
`type(sth)` → int

3) `b = 7.345`
`type(b)` → float



$$\left. \begin{aligned} x == y &\Leftrightarrow y \in (x - \varepsilon, x + \varepsilon) \\ (x - y) &< \varepsilon \end{aligned} \right\} \text{relative to } \varepsilon$$

`b = -3.45`

`abs(b)`

`c = -3.44444449`

`abs(c - b) < 0.0000001` → False

! In Python, comparăm 2 float-uri cu assert.

`assert (abs(c - b) < 0.01)`

~ gradienti

~ integrala dublă

~ derivata parțială.